6. Applications of Integrals

6.3. Volumes by Cylindrical Shells

6.3 Learning Objectives:
- I can use shell method to find the volume of solids of rotation.
- I can identify scenarios when shell or washer method is required for a problem based on its geometry.

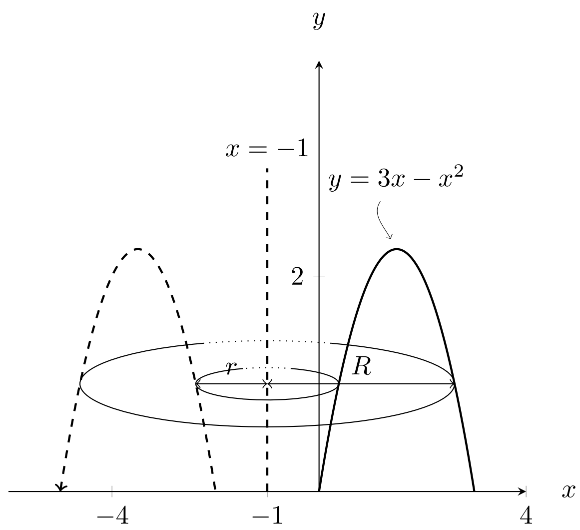
Exercise 6.3.1

Spend 5-10 minutes trying to set up the integral for using washer method for the following problem. Then stop and analyze why this is particularly difficult.

Rotate the region enclosed by the x-axis and the parabola $y = 3x - x^2$ around the line $x = -1$.

Solution

Note the plot of this region below



helps illustrate the difficulty. Simply put: We have no easy way of determining the radii, r and R.

In the disk and washer method, the cross-sections taken were circular and perpendicular to the axis of rotation. We will introduce a new method, using a new shape for cross-sections, which are taken parallel to the axis of rotation.

Consider the following cookie cutters.

For the next method, we will think of finding the area of a concentric shells making up the shape, then integrating from the center of the shells to the edge to “add” up all the areas that go into the volume.

6.3 Section Summary: